Yu Fan (Wesley) Chou

+65 8173 1981 | yufanchou583@gmail.com | github.com/noradar123 | linkedin.com/in/yu-fan-chou

Education

Nanyang Technological University Singapore
B.Sc. in Mathematical and Computer Sciences (Double Major), GPA 4.48/5.00 *Aug 2024 – May 2028*
Relevant coursework: Data Structures & Algorithms, Distributed Systems, Machine Learning, Neural Networks, Probability, Real Analysis.

Experience

Vernus AI — Research Intern Taipei, Taiwan
Python, scikit-learn, pandas *Jul 2026 – Present*
- Building a regression model that predicts chromatography (HPLC) retention from molecular descriptors, comparing descriptor sets and feature-selection methods against a ridge regression baseline.

Projects

Distributed Banking System SC4051, 2026
C++17, Go, UDP sockets
- Co-developed a C++17 client and Go UDP server supporting seven banking services, including fund transfers and update callbacks, over a hand-marshalled binary wire protocol.
- Implemented client-side retries with server-side reply caching to provide both at-least-once and at-most-once invocation semantics over an unreliable transport.
- Tested against duplicate transactions, callback expiry, and recovery from simulated reply loss, using a configurable packet-drop layer to reproduce failures deterministically.

Contact-Exchange Smartwatch App 2026
HarmonyOS, ArkTS *Huawei Wearable App Challenge (NTU) — 3rd Place*
- Built a wearable app that captures a new contact by scanning them in person, removing the need to look the person up on a separate platform during the conversation.
- Sole developer of the watch client: designed the on-device interaction flow and implemented the full front end under the constraints of a small round display and short interaction windows.

Robot Navigation: RRT Planning with Safety Constraints Sep 2025 – May 2026
Python, PyBullet *NTU (URECA) | Supervisor: Asst Prof Sung Yoonchang*
- Integrated RRT path planning with a Control Barrier Function quadratic-programming controller for navigation around dynamic obstacles, simulated in PyBullet.
- Derived second-order barrier constraints to modify acceleration commands, handling cases where replanning alone could not prevent collisions under inertial motion.
- Documented the failure modes of nearest-obstacle tracking and proposed an extension to multiple simultaneous constraints.

Momentum Strategy Backtesting Framework GitHub
Python, NumPy, pandas
- Built a vectorised backtesting engine with pluggable position-sizing and transaction-cost models, and a regression test suite covering signal causality and execution timing.

Technical Skills

Languages: Python, C++, C, Java, Go, SQL
Libraries and tools: NumPy, pandas, scikit-learn, PyTorch, Matplotlib, pytest, Git, Linux, LaTeX

Honors

- **International Chemistry Olympiad — Gold Medal (2023):** ranked 12th globally out of 348 contestants from 89 countries, on the four-person team for Chinese Taipei.
- **National Olympiads (Taiwan):** Geography, 12th nationally; Linguistics, top 30 nationally.